LINEAR REGRESSION ANALYSIS PLANNING AND REPORT

Predicting Medical Insurance Charges: PART 1 POE

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1. Introduction

In today's data-driven world, organisations increasingly rely on predictive analytics to guide decision-making processes. Within the healthcare insurance industry, accurately estimating medical costs is critical for pricing strategies, risk management, and long-term sustainability. This project focuses on developing a Linear Regression model to predict medical insurance charges based on demographic and lifestyle variables.

As a student transitioning from theoretical learning into practical application, this project represents an opportunity to bridge the gap between academic concepts and real-world data science practice. The aim is not only to build a predictive model but also to demonstrate a clear understanding of the analytical process, from planning and data exploration to model evaluation and improvement.

In an increasingly data-driven healthcare environment, predicting medical costs is essential. This report develops a Linear Regression model to estimate insurance charges using demographic and lifestyle data. This project reflects both technical and analytical skills expected from an Advanced Diploma graduate.

2. Methodology and Planning

A structured process was followed: EDA, preprocessing, feature selection, modelling, evaluation, and improvement. This ensured a systematic and reproducible workflow aligned with best practices.

A structured methodology was followed to ensure the analysis was both systematic and effective. The workflow consisted of:

- Dataset understanding
- Exploratory Data Analysis (EDA)
- Data preprocessing
- Feature selection
- Model training
- Model evaluation
- Residual analysis
- Model improvement

This step-by-step approach ensures transparency and reproducibility, which are essential in data science (Tukey, 1977). Planning the analysis beforehand also reduces the likelihood of errors and ensures alignment with the objectives of the project.

3. Dataset Overview

The dataset contains variables such as age, BMI, smoking status, region, and charges. The target variable is continuous, making it suitable for Linear Regression.

The dataset used in this analysis contains 1338 observations and includes the following variables:

- Age
- Sex
- BMI (Body Mass Index)
- Number of children
- Smoking status
- Region
- Medical charges (target variable)

The dependent variable, medical charges, is continuous in nature, making it appropriate for Linear Regression modelling.

From a theoretical perspective, Linear Regression is suitable when:

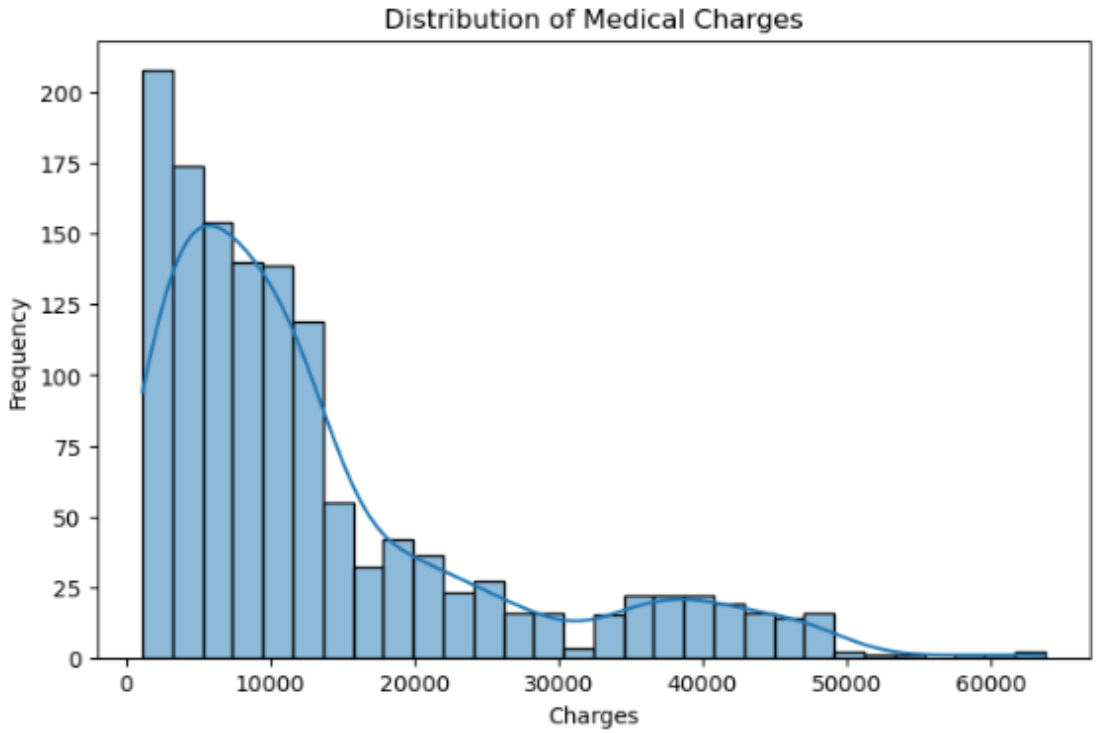
- The dependent variable is continuous
- Independent variables have a linear relationship with the target
- Residuals are approximately normally distributed

The dataset meets these conditions, and the variables included are logically linked to healthcare costs. For example, smoking and BMI are well-known contributors to increased medical risk, supporting the relevance of this dataset.

4. Exploratory Data Analysis

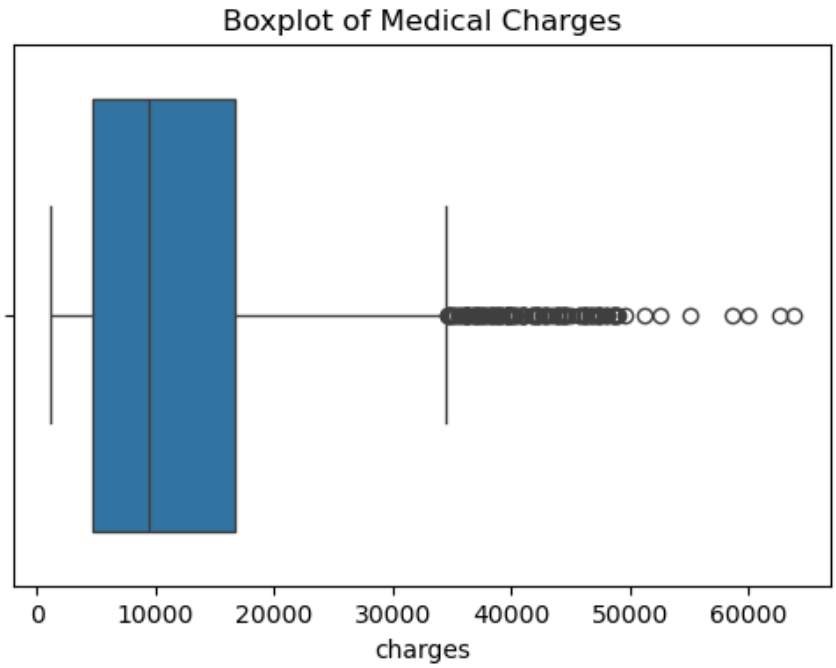
Exploratory Data Analysis was conducted to gain insights into the structure and patterns within the dataset.

4.1 Distribution of Charges



The distribution of medical charges is right-skewed. This indicates that the majority of individuals incur relatively low medical expenses, while a smaller group experiences significantly higher costs. This skewness suggests the presence of outliers, which may impact model performance.

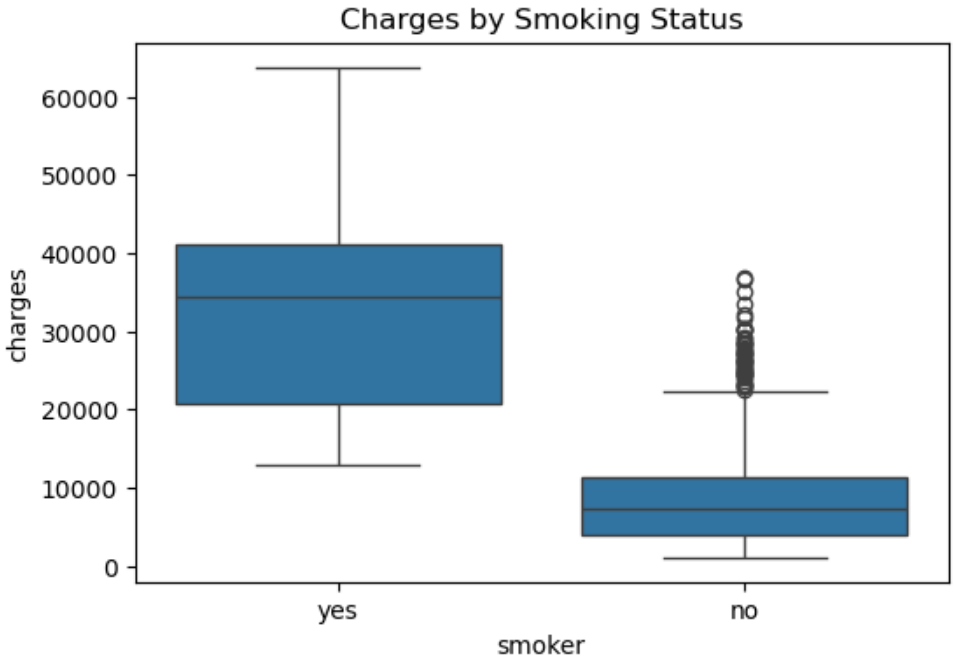
4.2 Outlier Detection



Outlier Detection

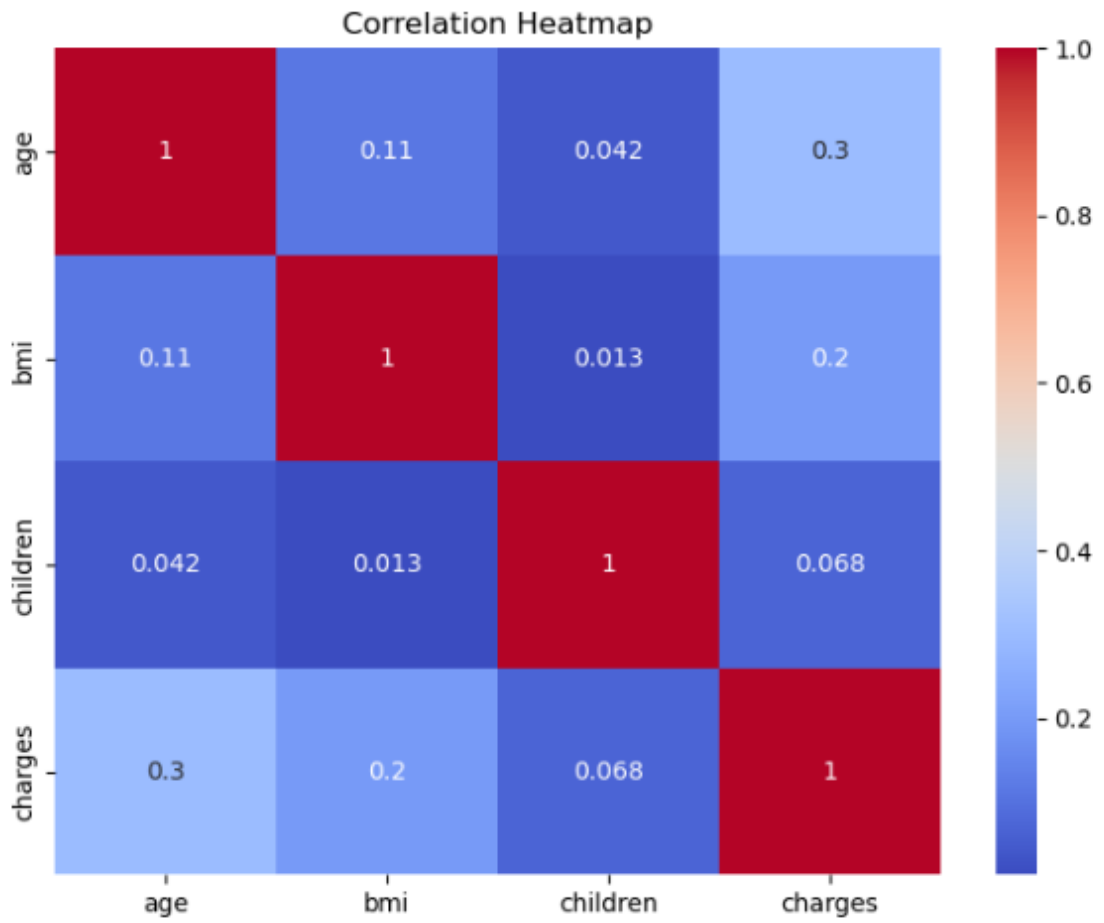
The boxplot confirms the presence of extreme values. These outliers represent individuals with unusually high medical expenses and may influence the regression model by increasing prediction error.

4.3 Smoking Status vs Charges



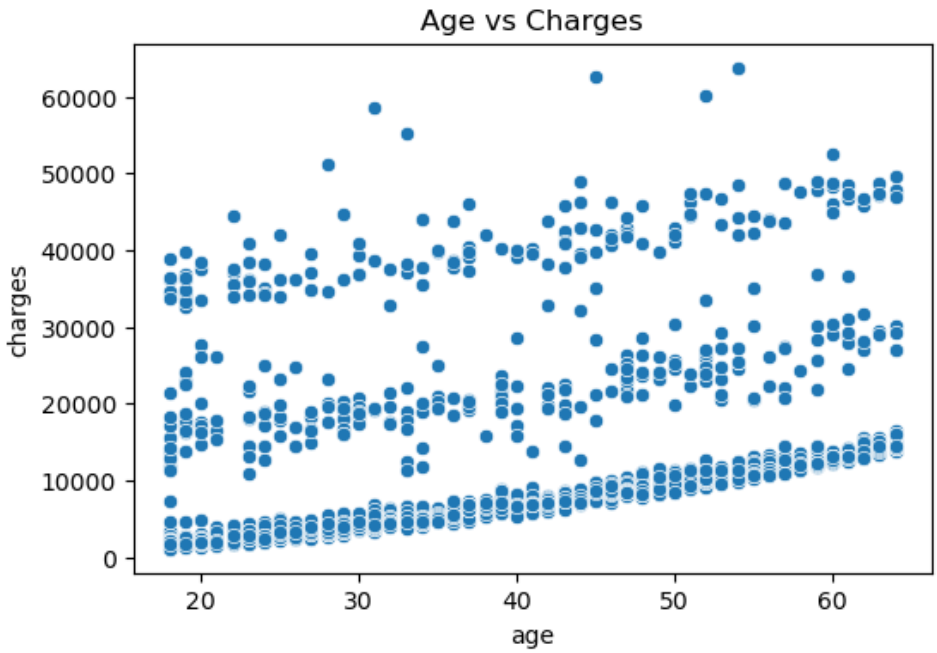
A clear distinction is observed between smokers and non-smokers. Smokers incur significantly higher medical costs, making this one of the strongest predictors in the dataset.

4.4 Correlation Analysis



The correlation heatmap shows moderate positive relationships between charges and variables such as age and BMI. This indicates that as these variables increase, medical costs tend to rise.

4.5 Age vs Charges



The scatter plot shows a positive trend, suggesting that older individuals generally incur higher medical expenses. This aligns with expectations, as healthcare needs typically increase with age.

EDA is a critical step in understanding the dataset before applying any modelling techniques.

The following steps will be performed:

- **Data Inspection**
  - Use .head(), .info(), and .describe() to understand structure and data types
- **Missing Values Check**
  - Identify and handle any missing values
- **Distribution Analysis**
  - Visualise the distribution of the target variable (charges) using histograms
- **Outlier Detection**
  - Use boxplots to identify extreme values
- **Relationship Analysis**

    - Use scatter plots and boxplots to analyse relationships between variables
- **Correlation Analysis**
    - Use a heatmap to identify relationships between numerical variables

5. Data Preprocessing

Before training the model, the dataset was prepared to ensure compatibility with machine learning algorithms.

Categorical variables were converted into numerical format using one-hot encoding. This step is necessary because machine learning models cannot interpret textual data directly (Pedregosa et al., 2011).

The dataset was then split into training and testing sets using an 80/20 ratio. This ensures that the model is evaluated on unseen data, providing a more accurate assessment of its performance.

Categorical variables were encoded using one-hot encoding. The dataset was split into training and testing sets.

6. Feature Selection

Feature selection was conducted using statistical methods, specifically p-values obtained from a regression model.

Variables such as age, BMI, number of children, and smoking status were found to be statistically significant predictors of medical charges. In contrast, some variables, such as gender and certain regional categories, were not significant.

Removing insignificant features helps improve model interpretability and reduces unnecessary complexity.

P-values were used to identify significant predictors. Smoking, age, and BMI were the strongest features.

7. Model Training

A Linear Regression model was trained using the Scikit-learn library. The model learned the relationship between the independent variables and the target variable.

The coefficients obtained from the model indicated that smoking status had the strongest impact on medical costs, followed by age and BMI.

Linear Regression model was trained using scikit-learn. The model learned relationships between features and charges.

8. Model Evaluation

The model was evaluated using three key metrics:

- R-squared: 0.7835
- Mean Absolute Error (MAE): 4185.81
- Root Mean Squared Error (RMSE): 5797.25

The R-squared value indicates that the model explains approximately 78.35% of the variation in medical charges. This suggests strong predictive performance.

The MAE indicates that, on average, predictions deviate from actual values by approximately 4185. This level of error is acceptable given the range of the data.

The RMSE is higher than the MAE, indicating the presence of larger errors, likely due to outliers identified during EDA.

R-squared: 0.7835 indicates strong performance. MAE and RMSE show acceptable prediction error.

	Feature	Coefficient
0	age	256.844160
1	bmi	336.613576
2	children	422.692610
3	smoker_yes	23657.580444
4	region_southeast	-474.760752
5	region_southwest	-627.124638

```
R-squared: 0.7835205571421147
Mean Absolute Error: 4185.809142538223
Root Mean Squared Error: 5797.2544275486625
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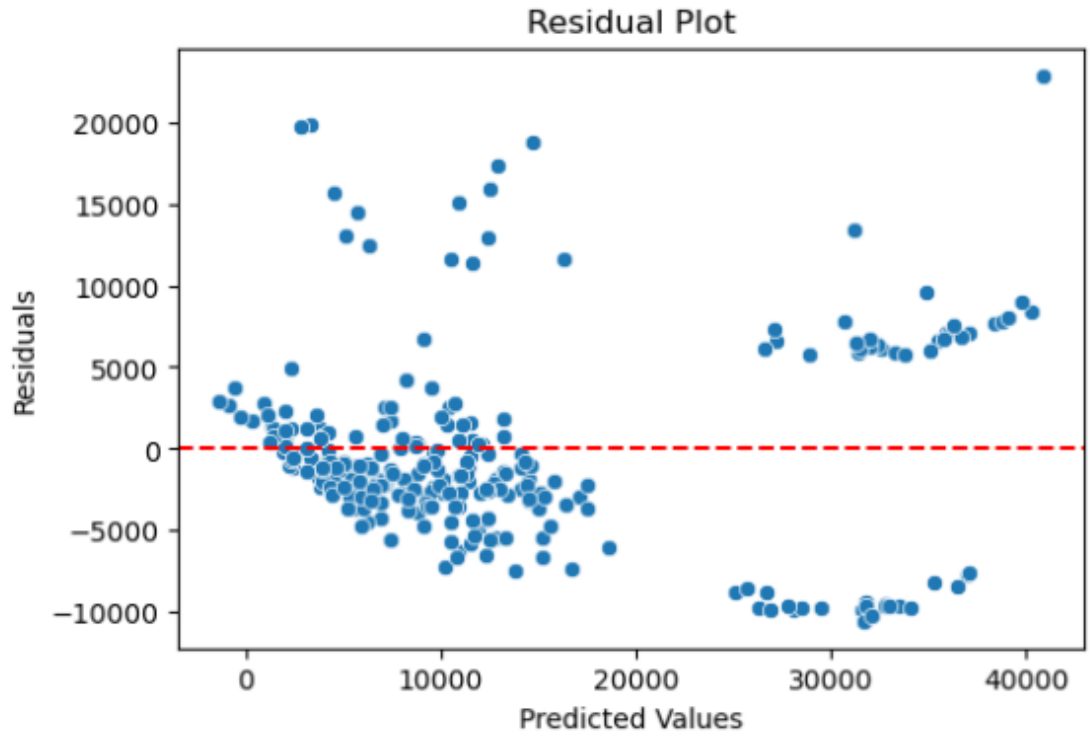
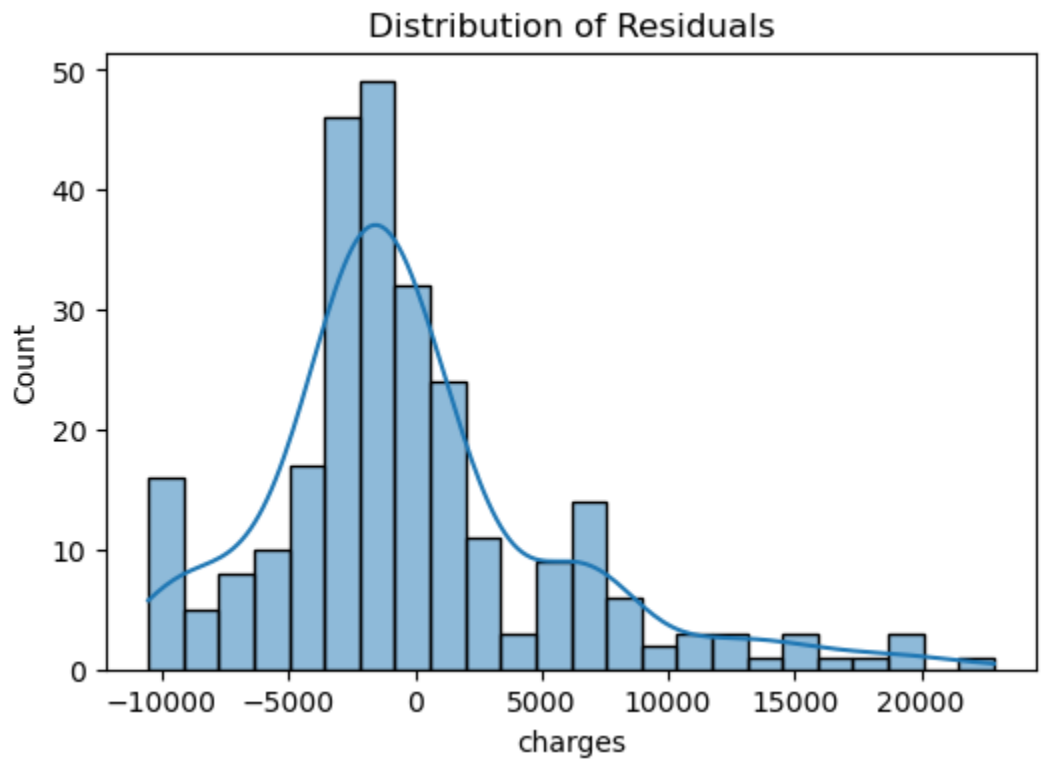
9. Residual Analysis





Residual analysis showed that errors are generally distributed around zero, indicating that the model captures the overall trend.

However, some spread is observed at higher values, suggesting that the model struggles slightly with extreme cases. This is consistent with earlier observations regarding outliers.

Residuals show reasonable distribution with slight variance at higher values.

10. Model Assumptions and Improvements

Linear Regression relies on several assumptions:

- Linearity
- Independence
- Homoscedasticity
- Normality of residuals

Based on the analysis conducted, these assumptions are reasonably satisfied, although minor deviations exist due to outliers.

Several potential improvements were identified:

- Applying Ridge or Lasso regression to reduce overfitting
- Transforming the target variable to reduce skewness
- Handling outliers more effectively
- Incorporating additional features

These improvements could enhance model performance and provide more accurate predictions.

Potential improvements include Ridge/Lasso and handling outliers.

11. Critical Reflection

From a learning perspective, this project provided valuable insight into the full data analysis pipeline. Initially, the focus was on understanding how to implement a model, but it became clear that the most important aspects are data preparation and interpretation.

One of the key challenges encountered was ensuring that the data was correctly formatted for statistical modelling. Resolving these issues helped deepen understanding of how machine learning models operate behind the scenes.

This project improved understanding of the full data pipeline. A key learning outcome was that data preparation is more important than the model itself.

12. Conclusion

This project successfully developed a Linear Regression model capable of predicting medical insurance charges based on demographic and lifestyle factors. The results demonstrate that variables such as smoking status, age, and BMI have a significant impact on medical costs. The model provides a reliable predictive tool and offers valuable insights that could be used by insurance providers to develop risk-based pricing strategies.

While the model performs well, further refinement could improve its ability to handle extreme values. The model effectively predicts medical costs. Smoking, age, and BMI are key predictors.

13. References

- Pedregosa, F. et al., 2011. *Scikit-learn: Machine Learning in Python*. Journal of Machine Learning Research.
- Tukey, J.W., 1977. *Exploratory Data Analysis*. Addison-Wesley.
- OpenAI, 2026. *ChatGPT (GPT-5.3)*. Available at: https://openai.com/ (Accessed: 26 March 2026).